THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Which of the following is NOT a metal? 1

(a) iron **(b)** copper **(c)** mercury **(d)** sulphur

2. A beam of light passes through a mixture and becomes visible. The mixture is a: 1

(a) true solution **(b)** pure compound **(c)** colloid **(d)** gas

3. A concave mirror of focal length 10 cm has a radius of curvature: 1

(a) 10 cm **(b)** 5 cm **(c)** 20 cm **(d)** 40 cm

4. A lunar year consists of 12 lunar months of 29.5 days each. The number of days is: 1

(a) 360 **(b)** 365 **(c)** 354 **(d)** 348

5. Which gas do aquatic plants release in the daytime that fish need? 1

(a) oxygen **(b)** carbon dioxide **(c)** methane **(d)** nitrogen

6. If air contains 21 per cent oxygen, in 500 L of air the volume of oxygen is about: 1

(a) 210 L **(b)** 78 L **(c)** 21 L **(d)** 105 L

7. A data table has more than one outlier. The best action is to: 1

(a) drop all of them **(b)** use the biggest

(c) check the experimental method **(d)** stop recording

8. A cell is stained in blue and the nucleus appears dark. The role of stain is to: 1

(a) kill the cell **(b)** increase size **(c)** provide contrast **(d)** make food

9. The immune system includes: 1

(a) white blood cells and antibodies **(b)** teeth and hair

(c) bones and nails **(d)** only skin colour

10. The magnetic effect of current is shown by a: 1

(a) wire melting **(b)** bulb glow

(c) bubble formation **(d)** compass needle deflection

11. Which of the following is NOT an effect of force? 1

(a) change in colour **(b)** change in shape

(c) change in direction **(d)** change in speed

12. A hurricane in the Atlantic and a typhoon in the Pacific are: 1

(a) different types of storms
(b) tornadoes
(c) earthquakes
(d) the same type of storm with different names

13. A gas at high pressure and low temperature is more likely to become: 1

(a) solid always **(b)** plasma **(c)** liquid **(d)** vapour only

14. A bell is made of bronze because it is: 1

(a) brittle **(b)** liquid
(c) sonorous and hard **(d)** soft

15. A student prepares tea with 5 g sugar in 100 mL and another with 10 g sugar in 100 mL. The second tea is: 1

(a) a suspension **(b)** more concentrated
(c) more dilute **(d)** saturated

16. A convex lens converges a parallel beam to a point called the: 1

(a) principal focus **(b)** axis **(c)** pole **(d)** optical centre

17. The Moon's waxing crescent is seen in the: 1

(a) south at noon **(b)** east at midnight
(c) west after sunset **(d)** north at dawn

18. In a pond, the amount of dissolved oxygen falls when: 1

(a) organic waste rots in it **(b)** it is cleaned
(c) plants grow **(d)** it rains

19. The percentage of Earth's water that is fresh water is about: 1

(a) 3 per cent **(b)** 97 per cent **(c)** 50 per cent **(d)** 30 per cent

20. In a pendulum test of the effect of the mass of the bob on the time period, which must be kept constant? 1

(a) number of masses **(b)** mass of the bob
(c) length of the string **(d)** time period

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Why does the volume of a mixture of alcohol and water appear less than the sum of their volumes? 2

22. Why is water called a compound and not a mixture? Give two reasons. 2

23. Find the mass percentage of a solution made by dissolving 18 g of sugar in 82 g of water. 2

24. A ray of light makes an angle of 30 degrees with a plane mirror. What is the angle between the incident and reflected rays? 2

25. A stick of height 2 m casts a shadow of 2 m at 10 am. What is the angle of the Sun above the horizon? 2

26. Why are there fewer top carnivores than herbivores in an ecosystem? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A solution is prepared by dissolving 35 g of salt in 140 g of water. Find the mass of the solution and the mass percentage of salt. 3

28. A ray falls on a plane mirror at 40 degrees to the normal. The mirror is turned by 10 degrees keeping the incident ray fixed. Find the new angle of incidence and the angle by which the reflected ray turns. 3

29. A student marks the tip of the shadow of a 1 m stick at 9 am, 12 noon and 3 pm. Describe the pattern and the shadow lengths relationship. 3

30. Construct a food chain of four organisms in a pond and identify the trophic level of each. 3

31. The Earth's radius is 6400 km. The crust is 35 km, mantle 2900 km. Find the total thickness of the core. 3

OR

A student measured the length of a table with a hand span and found 8; another got 9. Explain the difference and suggest a better method. 3

32. A cell is magnified 200x and measures 4 mm in the picture. Find its actual size in micrometres. 3

OR

Differentiate between first aid and medical treatment. Give an example of first aid. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Describe an experiment to verify the laws of reflection. State the apparatus, procedure, observations and conclusion. 5

34. Describe an activity to find the time of local noon using the shadow of a stick and explain how you find north-south. 5

35. A grassland has grass, grasshoppers, mice, frogs, snakes and hawks. Draw the food web in words, list the chains, and predict the effect of removing the snakes. 5

36. Describe the structure of the Earth's interior with the thickness (approximately), state and composition of each layer, and how it is studied. 5

OR

A report states 'The more fertiliser we gave the taller the plants got.' Write the variables, the likely hypothesis and ways to test it fairly. 5

37. What are the features of viruses? Mention two diseases caused by them and two ways of prevention. 5

OR

A family drinks river water directly. Explain the risks and a solution. 5

38. Describe the safety systems in a home electrical circuit: fuse, MCB, earthing, insulation, switches, and say what each protects against. 5

OR

Discuss the effect of air resistance on a falling feather, a stone and a parachutist. 5

END OF QUESTION PAPER — Total: 80 Marks